

Homework

1. Complete the following:
 - a. Write in tabular form:
 - i. $A = \{x: x \text{ is a letter in "do not repeat"}\}$
 - ii. $3\mathbb{Z} = \{x \in \mathbb{Z}: x = 3n \text{ for some } n \in \mathbb{Z}\}$
 - iii. $N = \{x: x \text{ is odd and } x \text{ is even}\}$ (recall that 0 is even)
 - b. Write these sets in set-builder form:
 - i. Let $A = \{3, 4, 5, 6, 7, \dots\}$
 - ii. Let $B = \{3, 5, 7, 9, \dots\}$
 - iii. Let $\mathbb{Q}$ be the set of rational numbers.
2. Find all sets of $A = \{1, 2, 3\}$.
3. Prove that every nonempty subset has a proper subset.
4. Prove that if A is a subset of $\emptyset$, then $A = \emptyset$.
5. Prove that $A = \{2, 3, 4, 5\}$ is not a subset of $B = \{x \mid x \text{ is even}\}$.
6. Prove that A and B are subsets of $A \cup B$.
7. Prove that $A = A \cup A$ and $A = A \cap A$.
8. Prove that $A \cap B$ is a subset of A and of B .
9. Prove that $A \cup B = \emptyset$ implies that $A = \emptyset$ and $B = \emptyset$.
10. Prove that $A \cup B = B$ if and only if $A \subseteq B$.
11. Prove that $A \cap B = B$ if and only if $B \subseteq A$.
12. Prove that $A \cup (A \cap B) = A$.
13. Prove that $A \cap (A \cup B) = A$.
14. Prove that $\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B)$.

Presentation problems.

1. Prove that $A - B$ is a subset of $A \cup B$.
2. Prove that if A a subset of B then $A \cap B = A$.
3. Let $A \cap B = \emptyset$. Prove that $B \cap A' = B$.
4. Prove that if A is a subset of B then $A \cup B = B$.
5. Prove that $A' - B' = B - A$.
6. Prove that if A is a subset of B then B' is a subset of A' .
7. Prove that if $A \cap B = \emptyset$ then $A \cup B' = B'$.
8. Prove that $(A \cap B)' = A' \cup B'$.
9. Prove that $A \subset B$ implies that $A \cup (B - A) = B$.